



Cambridge International AS & A Level

PHYSICS

9702/51

Paper 5 Planning, Analysis and Evaluation

May/June 2025

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance
For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	benefit of the doubt given
	correct awarding one mark from additional detail 1. similar numbered ticks are used for additional detail 2, 3, 4 etc.
	correct point or mark awarded
	defining the problem mark
	error carried forward applied
	error in number of significant figures
	incorrect or insufficient point ignored while marking the rest of the response
	incorrect point or mark not awarded
	incorrect unit
	information missing or insufficient for credit

Annotation	Meaning
MO	methods of data collection mark
SEEN	point has been noted, but no credit has been given or blank page seen
R	repeat of point previously awarded mark

Question	Answer	Marks
1	Defining the problem	
	vary f and measure V or f is the independent variable and V is the dependent variable	1
	keep E <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: - circuit with a.c. supply - oscilloscope connected in parallel with the resistor - workable circuit - oscilloscope and a.c. supply labelled	1
	labelled signal generator or <u>variable frequency</u> power supply connected across the terminals	1
	method to determine V or E from oscilloscope, e.g. multiply amplitude / height of wave by y -gain on oscilloscope	1
	method to determine f from oscilloscope, e.g. determine period T by multiplying number of divisions in 1 cycle or horizontal distance in 1 cycle by the time base and $f = 1/T$	1
	Method of Analysis	
	plot a graph of $\frac{1}{V}$ against f or equivalent, e.g. f against $\frac{1}{V}$ Allow logarithms e.g. $\lg V$ against $\lg f$.	1
	relationship valid <u>if</u> a straight line is produced <u>passing through the origin</u> (for $\lg V$ against $\lg f$: relationship valid <u>if</u> a straight line is produced with <u>gradient</u> = -1)	1

Question	Answer	Marks				
1	<table><tr><td>$\frac{1}{V}$ against f</td><td>f against $\frac{1}{V}$</td></tr><tr><td>$K = \frac{lES}{AN^2} \times \text{gradient}$</td><td>$K = \frac{lES}{AN^2} \times \frac{1}{\text{gradient}}$</td></tr></table> (for $\lg V$ against $\lg f$: $K = \frac{ElS}{AN^2} \times 10^{-y\text{-intercept}}$).	$\frac{1}{V}$ against f	f against $\frac{1}{V}$	$K = \frac{lES}{AN^2} \times \text{gradient}$	$K = \frac{lES}{AN^2} \times \frac{1}{\text{gradient}}$	1
$\frac{1}{V}$ against f	f against $\frac{1}{V}$					
$K = \frac{lES}{AN^2} \times \text{gradient}$	$K = \frac{lES}{AN^2} \times \frac{1}{\text{gradient}}$					
	Additional detail including safety considerations	6				
D1	precaution linked to <u>hot coil</u> or <u>hot resistor</u> or prevention of <u>burns</u> from <u>coil</u> or <u>resistor</u> , e.g. use gloves / switch off power supply when not measuring V to prevent burns from coil / resistor					
D2	keep N and A and l and S <u>constant</u>					
D3	method to keep S constant, e.g. switch off power supply between readings to prevent heating of resistor or to allow resistor to cool					
D4	method to determine A , e.g. use calipers / micrometer to measure diameter (of coil) / d and $A = \frac{\pi d^2}{4}$					
D5	repeat measurements of diameter d along the length of the coil / in different directions and determine the average value of d					
D6	method to determine the value of S , e.g. separate circuit diagram showing resistor connected to ohmmeter, or circuit diagram showing resistor connected to a power supply with an ammeter and voltmeter and $S = V / I$					
D7	measure l with a ruler / calipers					
D8	oscilloscope drawn connected across terminals / across signal generator and description to determine E					
D9	adjust y -gain for maximum amplitude or adjust time base for length of one wave or measure n waves and divide measured time by n					

Question	Answer	Marks
1	D10 method to keep E constant, e.g. check p.d. and alter supply or method to keep l constant, e.g. tape coil or method to keep A constant, e.g. wind wire on a cylinder	

Question	Answer	Marks							
2(a)	gradient = $\frac{R}{E}$ y-intercept = $\frac{Z}{E}$	1							
2(b)	<table><tr><td>$\frac{1}{I} / 10^3 \text{ A}^{-1}$</td></tr><tr><td>2.20 or 2.198</td></tr><tr><td>1.90 or 1.905</td></tr><tr><td>1.72 or 1.724</td></tr><tr><td>1.57 or 1.575</td></tr><tr><td>1.46 or 1.460</td></tr><tr><td>1.31 or 1.307</td></tr></table> Values of $\frac{1}{I} / 10^3 \text{ A}^{-1}$ correct as shown above.	$\frac{1}{I} / 10^3 \text{ A}^{-1}$	2.20 or 2.198	1.90 or 1.905	1.72 or 1.724	1.57 or 1.575	1.46 or 1.460	1.31 or 1.307	1
$\frac{1}{I} / 10^3 \text{ A}^{-1}$									
2.20 or 2.198									
1.90 or 1.905									
1.72 or 1.724									
1.57 or 1.575									
1.46 or 1.460									
1.31 or 1.307									

Question	Answer	Marks
2(b)	Uncertainties in $\frac{1}{I} / 10^3 \text{ A}^{-1}$ from ± 0.02 or ± 0.03 decreasing to ± 0.01 .	1
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1
	Error bars in $\frac{1}{I}$ plotted correctly. All error bars to be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	1
2(c)(ii)	Straight line of best fit drawn. Thickness of the line must be less than half a small square. Do not accept line from top point to bottom point. Line must pass between (0.101, 1.40) and (0.104, 1.40) and between (0.189, 2.10) and (0.194, 2.10)	1
	Worst acceptable straight line drawn (steepest or shallowest possible line that passes through all the error bars). Thickness of the line must be less than half a small square. All error bars must be plotted.	1
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient determined of worst acceptable line with clear substitution of data points into $\Delta y / \Delta x$. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1

Question	Answer	Marks
2(c)(iv)	y-intercept determined by substitution of correct point with consistent power of ten in m and y into $y = mx + c$.	1
	y-intercept of worst acceptable line determined by substitution into $y = mx + c$. uncertainty = y-intercept of line of best fit – y-intercept of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line y-intercept – shallowest worst line y-intercept) Do not accept ECF from false origin method.	1
2(d)(i)	R determined using gradient and R and Z given to 2 or 3 significant figures. $R = \text{gradient} \times 5.8$	1
	Z determined using y-intercept and R and Z given with units with appropriate powers of ten. $Z = \text{y-intercept} \times 5.8$ unit of R: Ω or V A^{-1} unit of Z: Ω or V A^{-1}	1
2(d)(ii)	Percentage uncertainty determined using $\Delta E = 0.2$ (V) with method shown. $\Delta R\% = \left(\frac{\Delta E}{E} + \frac{\Delta \text{gradient}}{\text{gradient}} \right) \times 100$ or $\Delta R\% = \left(\frac{0.2}{5.8} + \frac{\Delta \text{gradient}}{\text{gradient}} \right) \times 100$	1

Question	Answer	Marks
2(e)	<i>I</i> determined to a minimum of 2 significant figures from (c)(iii) and (c)(iv) or (d)(i) with correct substitution. $I = \frac{1}{\frac{\text{gradient}}{20} + y\text{-intercept}}$ or $I = \frac{E}{\left(\frac{R}{20} + Z\right)}$	1